

AI Social Media Content Calendar Generator – Hackathon Problem Brief

1. Problem Context

Content creators and small business owners face a recurring bottleneck: coming up with fresh, platform-specific post ideas week after week. They spend hours brainstorming captions, finding relevant hashtags, and reformatting the same idea for Instagram, LinkedIn, and X — which have vastly different audiences and tone requirements. Existing tools like Buffer or Later handle scheduling but don't generate ideas; Canva focuses on design but not strategy. There's a gap for an intelligent, all-in-one tool that removes the creative block and adapts content automatically across platforms.

2. Objective

Build a web application where users input their brand identity and posting goals, and an AI instantly generates a month-long, platform-specific content calendar with post ideas, captions, and hashtag suggestions. Users should be able to regenerate individual posts, drag-and-drop to reschedule, and export their calendar for use or sharing.

3. Core Features (Must-Have)

- **User Authentication & Brand Profile:** Sign up/login, and create a profile with brand name, niche/industry, target audience, and tone (e.g., professional, playful, inspirational).
- **AI Content Calendar Generation:** Input posting goals (e.g., "3 posts per week on product launches, behind-the-scenes, and tips"), and generate 30 posts with custom ideas, captions, and hashtags in one click.
- **Platform-Specific Adaptation:** Display content tailored for Instagram (emoji-heavy, conversational), LinkedIn (professional, value-focused), and X (concise, punchy) — same core idea, different voice for each.
- **Individual Post Regeneration:** Click a "refresh" button on any post to get an alternative idea/caption without regenerating the entire calendar.
- **Drag-and-Drop Calendar View:** Visual month-long calendar where users drag posts to different dates to reschedule.
- **Export/Download:** Save the calendar as PDF, CSV, or JSON for external use or sharing with the team.
- **Post Editing:** Allow users to manually edit any AI-generated caption, hashtags, or ideas before finalizing.

4. Stretch Features (Good-to-Have)

- **Image Suggestions:** Integrate Unsplash or Pexels API to suggest images for each post based on its topic.
- **Performance Analytics Dashboard:** Show mock analytics on past posts (e.g., engagement, impressions) to inform future content strategy.
- **Multi-User Collaboration:** Allow team members to join a brand workspace, comment on posts, and approve calendar changes.

- **Direct Posting Integration:** OAuth connect to Instagram/LinkedIn and auto-post directly from the app (or mock this for the demo).

5. Suggested Tech Stack

Frontend: React.js (with TypeScript optional) – Fast, component-based, and has excellent drag-and-drop libraries; easy for beginners to learn and quick to build interactive UIs.

Backend: Node.js + Express – Lightweight, non-blocking, and JavaScript throughout simplifies the stack; fast to set up REST APIs.

Database: Firebase Firestore or MongoDB Atlas – Flexible schema for storing user profiles, brands, posts, and calendars; Firebase includes free authentication and hosting, reducing setup time.

AI/Content Generation: OpenAI API (GPT-4 or GPT-4o) or Anthropic Claude API – Pre-trained models, easy to call via REST, and free tier available for hackathon testing; no need to train your own model.

Drag-and-Drop: react-beautiful-dnd or React DnD – Mature, easy-to-use libraries specifically built for drag-and-drop in React.

Deployment: Vercel (frontend) + Render/Railway (backend) – Both have free tiers, simple Git-based deployments, and work well for a 24-hour sprint.

6. Suggested Approach / Step-by-Step Plan

1. **Project Setup (30 min):** Initialize React frontend, Node/Express backend, and connect to a Firebase project. Set up environment variables for API keys.
2. **Design Database Schema (20 min):** Plan collections/tables for Users, Brands, ContentPosts, and Calendars. Keep it simple: User → Brand → Calendar → Posts.
3. **Build Authentication Flow (45 min):** Implement Firebase sign-up/login on frontend; create protected routes on backend that check user ownership before returning data.
4. **Create Brand Profile Form (30 min):** Build a simple form to capture brand name, niche, tone, audience, and posting goals. Save to database.
5. **Implement Calendar Generation API (1–1.5 hours):** Write a backend endpoint `/generate-calendar` that takes brand info, constructs a detailed prompt, calls OpenAI API to generate 30 posts (4–5 per platform), and stores them. Test with mock data first.
6. **Build Calendar UI (1 hour):** Create a React component that displays posts in a month grid (e.g., using a simple table or react-calendar). Show platform tags, captions, and hashtags in a readable format.
7. **Add Drag-and-Drop (45 min):** Integrate react-beautiful-dnd; enable users to drag posts between dates. Update the backend when a post is rescheduled.

8. **Implement Post Regeneration (30 min):** Add a "refresh" button on each post that calls `/regenerate-post` endpoint. Fetch a new idea for just that post without touching others.
9. **Build Export Functionality (30 min):** Add a button to download the calendar as PDF (use jsPDF or similar) or CSV.
10. **Polish, Test & Deploy (1 hour):** Test the full flow end-to-end, fix bugs, deploy frontend to Vercel and backend to Render, and prepare a live demo.

7. Sample Data / APIs to Use

OpenAI API (<https://platform.openai.com/>) – Call with a prompt like: "Generate 5 Instagram posts for a sustainable fashion brand. Each post should include a caption (100 words max), 3 relevant hashtags, and a post type (product launch, behind-the-scenes, or educational tip). Format as JSON."

Alternative: Anthropic Claude API (<https://www.anthropic.com/product>) – Similar usage; Claude is good at generating varied, natural-sounding content.

Unsplash API (<https://unsplash.com/developers>) – Free image search API; call it with keywords extracted from post topics to fetch relevant images (stretch feature).

Firebase (<https://firebase.google.com/>) – Use Firestore for real-time database, Firebase Auth for user management, and Hosting for deployment.

Mock Data for Testing: Before calling the API, create a JSON file with 3–4 sample posts per platform. This lets you test the UI and drag-and-drop without burning API credits.

8. Key Challenges to Anticipate

Challenge 1: AI API Costs and Rate Limiting

During a hackathon, repeated calls to OpenAI can add up. **Tip:** Use OpenAI's free trial credits or set a low cost cap. Cache generated calendars so regenerating a post doesn't regenerate the entire month. Batch 4–5 post generation calls into one API request rather than calling once per post.

Challenge 2: Drag-and-Drop Complexity

Implementing smooth, responsive drag-and-drop is trickier than it seems, especially with date logic and state management. **Tip:** Use a proven library like `react-beautiful-dnd` and start with a simple grid layout. Test dragging between a few dates before handling edge cases like dragging across weeks.

Challenge 3: Platform-Specific Content Quality

Generating genuinely good captions that differ meaningfully across platforms requires careful prompt engineering. **Tip:** Write detailed platform-specific prompts upfront (e.g., "For Instagram: use emojis, conversational tone, call-to-action"; "For LinkedIn: professional, value-driven, industry insights"). A few well-crafted examples in the prompt help the AI understand expectations.

Challenge 4: User Experience Overwhelm

Showing 30 posts, multiple platforms, and editing options can feel cluttered. **Tip:** Default to a clean month-grid view and hide details (captions, hashtags) until a user clicks a post. Add a filter to show only one platform at a time.

9. What a Good Demo Looks Like

A compelling demo shows a live walkthrough: A judge signs up, creates a brand profile (e.g., "eco-friendly yoga studio"), specifies posting goals ("2 yoga tips, 1 event promotion, 1 community story per week"), and clicks "Generate Calendar." Within seconds, a 30-post calendar appears, color-coded by platform. The judge sees Instagram posts with emojis and engagement-focused copy, LinkedIn posts with professional insights, and X posts that are snappy and link-friendly. The judge then drags a post to a different date to show scheduling works, clicks "refresh" on one post to see an alternative idea appear instantly, and exports the calendar as PDF. The demo ends with the exported file open, showing a clean, ready-to-use content roadmap. Throughout, the UI is intuitive, loading is fast, and the AI content feels relevant and varied — not templated or repetitive.